Villa Somalia SCADA Design Report

Rev 1 · 3 Nov 2026 · @Eli

1. Introduction

This report sets out the SCADA design for the Villa Somalia hybrid power plant and is submitted for the client's approval. It answers the client's review comments on the earlier drawing set.

Revision 1 confirms Siemens SICAM A8000 as the selected architecture, in reply to the Employer's comment on RFA-001/BRA/26 Issue H. It is the one coordinated design package for review and approval. It supersedes the original submission (Rev 0), and the alternative submitted on Schneider Electric Modicon M580 is withdrawn in full. The design is that of the original submission; the changes are listed in the revision history below.

Project

The plant supplies the Office of the President and the Office of the Prime Minister in Mogadishu. It combines a 1,516 kWp PV array, a 3,340 kWh battery and two diesel generators on an 11 kV switchboard, with no utility grid connection.

Scope of the SCADA

The SCADA monitors every part of the plant on one system and lets authorised users switch breakers and change selected settings. It covers:

- PV inverters, battery, generators and the embedded energy management system (EMS)
- The 11 kV switchboard, the two office transformer stations and their LV distribution boards
- Transformers, auxiliary supplies, UPS, weather station, and fire detection and suppression

Three functions stay outside the SCADA and keep working if it fails:

1. **Dispatch** of PV, battery and generators stays with the EMS.
2. **Electrical protection** stays with the protection relays.
3. **Emergency stop and fire trips** are hardwired and act directly on the equipment.

Reference documents

Document	Content
Drawing set VILLA_SOMALIA_05_10_2026	Layouts, single-line diagram, communication schematic, civil and earthing drawings
Client review comments	SCADA, I/O list, communication, cybersecurity and datasheet requirements
Villa_Somalia_SCADA_Device_and_IO_List_Rev1.	Device list, protocols, I/O list, I/O rows,

Document	Content
xlsx	communication points (Attachment B)
Drawing set BFE/PR/26/01-SCADA-001 to 005, Rev B	SCADA architecture, communication network, PV interface, BESS interface, SCADA overview SLD (Attachment A)
Villa_Somalia_SCADA_Attachment_C_HMI_Screens_Rev1.pdf	HMI screen mock-ups, 12 screens (Attachment C)
Villa_Somalia_SCADA_Bill_of_Material_Rev1.xlsx	Bill of material by panel and by order line (Attachment D)
RFA-001/BRA/26 Issue H, Employer's comment of 20 Oct 2026	Confirm the selected architecture and resubmit one coordinated package

Revision history

Revision	Description and status
Rev 0, October 2026	Original submission on Siemens SICAM A8000 (its documents are headed "Rev 1"). Superseded by Rev 1.
Alternative, October 2026	Alternative on Schneider Electric Modicon M580 and Machine SCADA Expert, submitted with the original. Withdrawn: not to be used for review, procurement or construction.
Rev 1, 3 Nov 2026	First revision. SICAM A8000 confirmed (RFA-001/BRA/26 Issue H). Design as the original submission. Attachments: A drawings BFE/PR/26/01-SCADA-001 to 005 Rev B, B device and I/O list, C HMI screen mock-ups, D bill of material. Server rack and SW-MV enclosure added to the equipment list. For approval.

2. Compliance table

Every client comment is answered in this report or its attachments.

No.	Client comment	Response	Section
1	Please submit for approval	This report with attachments A to D	All
2	Provide complete SCADA design aligned	A redundant controller pair and one SCADA	3, 4, 5, 6, 12,

No.	Client comment	Response	Section
	with the BESS and PV side and the rest of the equipment, including the GIS combiner panel, LVDBs and generators	covering every equipment group, with screen mock-ups	Attachment C
3	Complete I/O list with all analogue and digital signals: VCB breakers, relays of the GIS combiner panel, inverter transformers, RMU, DGs, combiner panels, weather station, UPS, auxiliary power, fire detection and suppression	244 hardwired signals and 463 communication points, listed signal by signal, including the PV inverter AC breakers and the battery combiner panels	7, Attachment B
4	Submit communication architecture, protocols, network topology, cybersecurity provisions, and integration with GIS, LVDBs, generators, PV and BESS	Protocol for every link, redundant fibre ring, five-zone security design	5, 6, 8, 9
5	Provide datasheets for PPC controllers, field devices, servers and HMI, and the operational sequence	Bill of materials with model numbers; datasheets from the suppliers; operating sequences	11, 13, Attachment D

3. Design basis

The design is based on the equipment in drawing set VILLA_SOMALIA_05_10_2026 and the assumptions below.

Plant equipment

Equipment	Rating or model	Quantity
PV array	2,106 modules x 720 Wp =	1

Equipment	Rating or model	Quantity
	1,516 kWp	
PV string inverters NB01 to NB04	Sungrow SG350HX, 350 kW	4
Power line communication box (PLC box)	Converts inverter data to Ethernet	1
Battery energy storage system	Sungrow ST3340UX, 3,340 kWh	1
Embedded energy management system	Supplied with the PV and battery package	1
Diesel generators DG1, DG2, with step-up transformer	DG1 550 kVA, DG2 660 kVA; step-up transformer 1.5 MVA, 11/0.4 kV	2
11 kV switchboard, 1250 A busbar	Panels AH1 to AH8	1
Office transformer stations: RMU with ST260E relay, 630 kVA transformer, 10-way LVDB	President and Prime Minister	2
Auxiliary transformer	100 kVA	1
Neutral earthing resistor (panel AH8)	63.5 ohm, 100 A, 10 s	1
Weather station	Kipp & Zonen SMP12-A, Gill WindSonic, temperature sensors, Meteocontrol blue'Log XM	1
Battery fire detection and suppression	Per battery supplier	1

11 kV switchboard panels

Panel	Function
AH1	Diesel generator incomer
AH2	PV incomer
AH3	Battery incomer
AH4	Feeder to the Office of the President

Panel	Function
AH5	Auxiliary transformer feeder
AH6	Feeder to the Office of the Prime Minister
AH7	Spare feeder
AH8	Neutral earthing resistor

Assumptions

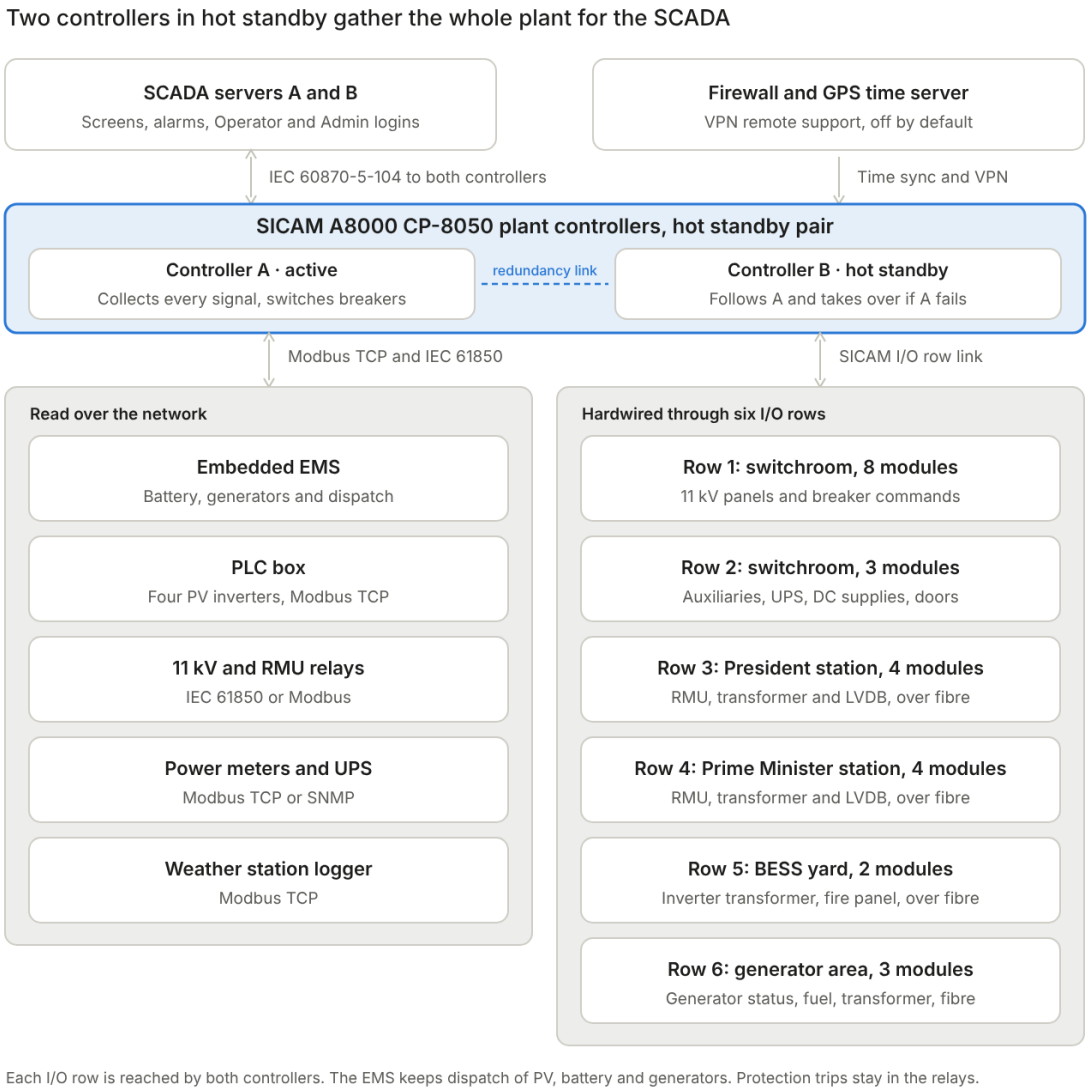
1. **Islanded plant.** The single-line diagram shows no utility incomer.
2. **Control voltage.** The switchgear auxiliary supply is 110 V DC. This sets the I/O module variants.
3. **Remote operation.** The 11 kV breakers, RMU switches and LVDB incomers are motorised.
4. **EMS interface.** The embedded EMS offers Modbus TCP to an external controller.
5. **PV data path.** Inverter data reaches the controller from the PLC box over Ethernet, as on the existing communication sheet.
6. **Relay protocol.** The 11 kV relays offer IEC 61850 or Modbus TCP.
7. **Signal counts.** Signals per device are typical values, to be checked against the terminal schedules.

Standards

Standard	Applied to
IEC 61850	Communication with the 11 kV protection relays
IEC 60870-5-104	Link between the controller and the SCADA servers
IEC 62443	Structure of the cybersecurity provisions
Modbus TCP and RTU	EMS, PLC box, generator controllers, meters, RMU relays, weather logger

4. System architecture

A hot standby pair of Siemens SICAM A8000 CP-8050 controllers collects every signal in the plant and passes it to two SCADA servers. Controller A is in control; controller B follows it and takes over if A fails. It reads five groups of devices over the network and six rows of hardwired signals.



SCADA architecture · redundant controller pair, 5 network sources, 6 I/O rows

Status and measurements flow up to the SCADA; breaker commands and permitted EMS settings flow down. A command passes three checks before a breaker moves: the SCADA checks the user's role, the controller checks interlocks and the local/remote switch, and the command module checks that only one output operates.

Components and their roles

Component	Role	Location
SICAM A8000 CP-8050, hot standby pair A and B	Plant controller: collects all signals, time-stamps events, executes breaker commands, passes settings to the EMS; B	11 kV switchroom, in the SCADA cabinet beside the switchboard

Component	Role	Location
	takes over if A fails	
SCADA servers A and B (SICAM SCC)	Screens, alarms, trends, reports, user logins and audit log; B takes over if A fails	Control room
Operator workstation	Day-to-day operation	Control room
Embedded EMS	Dispatch of PV, battery and generators	BESS area
Protection relays	Protection of the 11 kV and RMU circuits	11 kV switchboard and office stations
I/O rows 1 to 6	Hardwired signals from the field	Near the equipment they serve

Division of duties

Function	Carried out by	If the SCADA or controller fails
Dispatch of PV, battery and generators	EMS	Continues
Electrical protection	Relays	Continues
Emergency stop and fire trips	Hardwired circuits	Continue
Monitoring, alarms, event record	Controller and SCADA	Continue on the standby controller or server; lost only if both fail
Remote breaker switching	Controller and SCADA	Continues on the standby; if both controllers fail, breakers are operated locally
Generator start if the EMS fails	Generator controllers, on dead busbar	Continues
Load shedding on under-frequency	Relays AH6 and AH4	Continues

5. Scope of monitoring and control

Each equipment group reaches the controller over a communication protocol, by hardwired contacts, or both. Critical states are hardwired as well as communicated, so they stay visible if a link fails.

Equipment	By protocol	Hardwired	Monitored	Controlled from SCADA
PV inverters NB01 to NB04	Modbus TCP from the PLC box	Main LV breaker and the four inverter AC breakers to row 5	Power, voltages, currents, energy, temperature, insulation, state, fault codes	None; output is set by the EMS
Embedded EMS	Modbus TCP	None	Plant power balance, operating mode, generator requests, alarms	Operating mode, state-of-charge limits, generator thresholds, battery power limit, C-rate limit (Admin only)
Battery ST3340UX	Through the EMS	Fire, E-stop, door and main LV breaker to row 5	State of charge and health, power, cell limits, charge and discharge limits, combiner panel status, alarms	Through the EMS only
Diesel generators DG1, DG2	Modbus from each generator controller	Breaker status, running, alarm, E-stop, fuel level and step-up transformer LV breaker to row 6	Electrical output, engine values, counters, warnings, shutdown causes	Start and stop in manual mode
11 kV switchboard AH1 to AH8 (the GIS combiner panel)	IEC 61850 or Modbus TCP from the relays	Breaker, earth switch, local/remote, spring, gas, relay healthy to row 1, including the AH8 breaker	Currents, voltages, power, energy, trips and events	Breaker open and close on AH1 to AH7
RMUs at the office stations	Modbus RTU from the ST260E relays, through gateways	Switch positions, trips, gas, local/remote and RMU UPS to rows 3 and 4	Currents, trips, switch positions	Ring switches and transformer breaker
LVDBs at the	Modbus TCP from	Incomer and 10	Voltage, current,	Incomer open and

Equipment	By protocol	Hardwired	Monitored	Controlled from SCADA
office stations	power meters	feeder statuses to rows 3 and 4	power, energy, harmonics	close
Transformers (inverter, DG step-up, auxiliary, 2 x 630 kVA)	None	Temperature, pressure and oil level contacts, and the auxiliary transformer fuse, to the nearest row	Alarms and trips	None
UPS and DC supplies	Modbus TCP or SNMP	Status contacts and DC voltages to row 2	Battery state, faults, bus voltages	None
Fire detection and suppression	None	Pre-alarm, fire, fault, release and isolation to row 5	All panel states	None; reset at the panel only
Weather station	Modbus TCP from the blue'Log	None	Irradiance, wind, ambient and module temperature	None

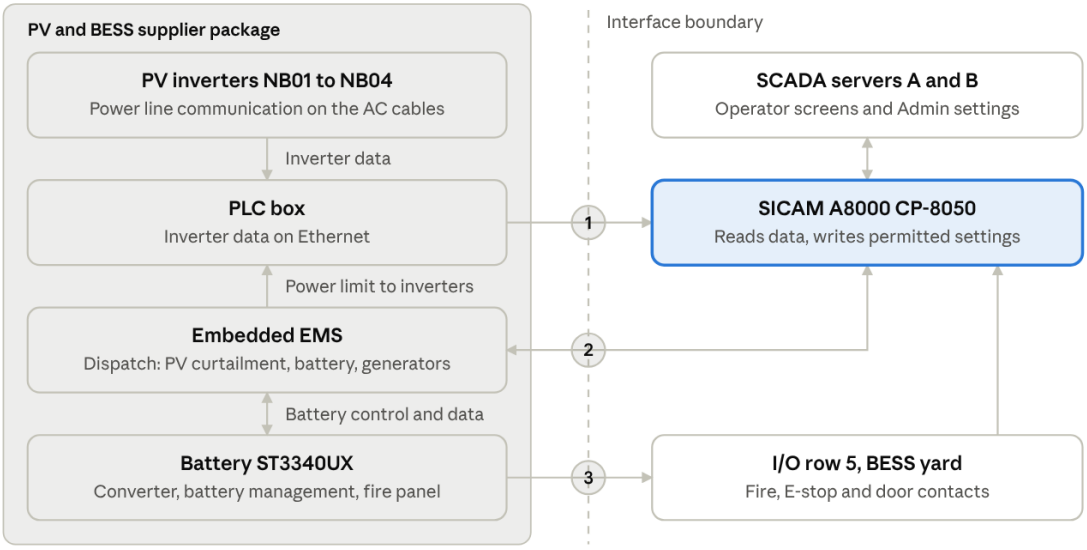
Integration with each system named by the client

- **GIS combiner panel.** Relays report over IEC 61850 or Modbus TCP; positions and commands are also hardwired to row 1.
- **LVDBs.** Power meters report over Modbus TCP; breaker positions are hardwired to rows 3 and 4.
- **Generators.** The generator controllers report over Modbus; critical states are duplicated by contacts to row 6.
- **PV.** Inverter data arrives from the PLC box over Modbus TCP.
- **BESS.** Battery data arrives through the EMS; safety states are hardwired to row 5.

6. Integration with the PV and BESS systems

The PV and battery arrive as one supplier package with its own EMS. The SCADA connects to that package at three defined points and never duplicates its control. This section fixes the boundary so both parties can design and test to it.

The SCADA meets the PV and BESS package at three defined points



Numbers mark the three interface points described in the table below.

PV and BESS interface · 3 interface points across the supplier boundary

Everything left of the boundary is designed, configured and warranted by the PV and BESS supplier. Everything right of it is the SCADA scope. The 11 kV incomers for PV (AH2) and battery (AH3) are on the SCADA side: they are monitored and switched through the relays and I/O row 1.

Interface points

No.	Interface	From and to	Protocol and medium	Direction	Content
1	PV data	PLC box to CP-8050	Modbus TCP, Ethernet on the fibre ring	Read only	Inverter measurements, energy, states, fault codes, link health
2	EMS	EMS and CP-8050, both ways	Modbus TCP, Ethernet on the fibre ring	Read; write permitted settings only	Read: plant power balance, battery data, generator requests, mode, alarms. Write: mode,

No.	Interface	From and to	Protocol and medium	Direction	Content
					state-of-charge limits, generator thresholds, battery power limit, C-rate limit, fault reset
3	Safety contacts	Battery fire panel, emergency stop, container door and the PV and battery LV breakers to I/O row 5	Dry contacts, hardwired	Read only	Fire panel states, emergency stop, door open, LV breaker positions and trips

Control authority

Each function has one place where it is executed and one route by which it may be changed. The SCADA never commands what the EMS controls; it changes EMS behaviour only through the EMS's own settings.

Function	Executed by	Changed by	SCADA role
PV output and curtailment	EMS, through the PLC box	EMS settings	Monitor
Battery charge and discharge	EMS and battery converter	Admin, through EMS setpoints	Monitor; Admin setpoints
Grid forming: voltage and frequency	Battery converter	Supplier settings only	Monitor
Battery C-rate limit for calibration	Battery, through the EMS	Admin only	Admin command, if the EMS exposes it
Generator start and stop, automatic	EMS, by hardwired start contact to the generator controllers	Admin, through EMS thresholds	Monitor
Generator start and stop, manual	Generator controller	Operator	Command

Function	Executed by	Changed by	SCADA role
11 kV breakers AH2 (PV) and AH3 (battery)	CP-8050 and secured command outputs	Operator	Command with interlocks
Protection settings	Relays	Protection engineer, locally	Monitor trips
Emergency stop and fire trip	Hardwired circuits	Nobody, remotely	Monitor
Load shedding on under-frequency	Relays AH6, then AH4	Protection engineer, locally	Monitor trips; reclose feeders

The EMS's own local screen remains available for supplier service, under a separate Admin password, so settings are not changed from two places in normal operation.

Interface signals

Source	Read points	Write points	Examples
PLC box, 4 inverters	About 88	8, used only if the EMS allows	Active and reactive power, AC and DC values, energy, temperature, insulation, fault codes
EMS	About 14	9	Plant power balance, mode, generator request, setpoints, fault reset
Battery, through the EMS	About 27	None	State of charge and health, cell limits, charge and discharge limits, alarm levels, combiner panel status
Hardwired, I/O row 5	18	None	Fire panel states, emergency stop, container door, LV breakers, inverter AC breakers

The point-by-point list is in Attachment B, sheets Comms Points and IO List. It will be replaced by the supplier's point tables when received.

Responsibilities at the interface

Item	PV and BESS supplier	SCADA contractor
Modbus point tables for the EMS and PLC box	Provides	Maps into the controller and SCADA
Settings that may be written remotely	Defines and confirms in writing	Implements, Admin role only
Network ports and IP addresses	Provides ports; agrees addresses	Issues the address and VLAN plan; provides cabling and switch ports
Time synchronisation	Configures NTP on the EMS and PLC box	Provides the GPS time server
Alarm list and alarm levels	Provides its alarm list	Maps it to levels 1 to 3
Fire, emergency stop and door contacts	Provides potential-free contacts at a terminal	Wires them to I/O row 5
Equipment names and tags	Uses the single-line diagram names (NB01 to NB04, AH1 to AH8, DG1, DG2)	Uses the same names on every screen and in the I/O list
Default passwords on the EMS and PLC box	Changes them and hands them over	Records them in the site register
Interface test	Attends, with the EMS in test mode	Leads and records

Interface test

Both parties test the interface together at commissioning and sign the results.

1. **Read points.** Each value on the SCADA matches the EMS or PLC box display.
2. **Write points.** Each permitted setting is changed by an Admin and confirmed at the EMS. Each forbidden write is confirmed blocked, and an Operator login is confirmed unable to write.
3. **Link loss.** Each link is disconnected; the SCADA raises an alarm, marks values stale and blocks commands. On reconnection it recovers without intervention.
4. **Safety contacts.** Each contact is operated at source and seen on the SCADA.
5. **Time stamps.** An event recorded by the EMS and by the SCADA shows the same time.

7. I/O summary

The plant has 244 hardwired signals on 24 I/O modules in six rows, with 27% spare channels. Row 5, in the BESS and inverter yard, is the fullest at 19% spare; a third input module can be added there if more yard signals appear. A further 463 signals arrive over communication links. The full signal-by-signal list is

in Attachment B: sheet IO List for hardwired signals and sheet Comms Point List for communication points.

I/O rows

Each row is a panel of I/O modules placed near the equipment it serves. Field contacts are wired a short distance to that panel, and one link carries all its signals to the controllers. Both controllers reach every row; only the active one reads inputs and drives outputs.

Row	Location	Interface to controller	Link	Equipment wired	Modules	Signals
1	11 kV switchroom, SCADA cabinet	CI-8532, electrical	Copper	11 kV panels AH1 to AH8, alarm beacon	8	87
2	11 kV switchroom, SCADA cabinet	CI-8532, electrical	Copper	Auxiliary transformer, UPS, DC supplies, switchroom E-stop and doors	3	25
3	President office station	CI-8533, optical	Fibre	RMU, 630 kVA transformer, LVDB, RMU UPS	4	43
4	Prime Minister office station	CI-8533, optical	Fibre	RMU, 630 kVA transformer, LVDB, RMU UPS	4	43
5	BESS and inverter transformer yard	CI-8533, optical	Fibre	Inverter transformer, PV and BESS LV breakers, inverter AC breakers, fire panel, BESS E-stop	2	26

Row	Location	Interface to controller	Link	Equipment wired	Modules	Signals
				and door		
6	Generator area	CI-8533, optical	Fibre	Generator status and fuel level, DG step-up transformer and its LV breaker	3	20
Total					**24**	**244**

The controller accepts up to 16 rows of 8 modules, so 10 rows remain free for expansion.

Modules

Module	Function	Quantity	Channels	Used	Spare
DI-8112	Digital input, 16 channels	17	272	208	64
DO-8221	Secured command output, 12 channels	4	48	30	18
DO-8212	Relay output, 8 channels	1	8	2	6
AI-8320	Analogue input, 4 channels	2	8	4	4
Total		**24**	**336**	**244**	**92**

Hardwired versus communicated

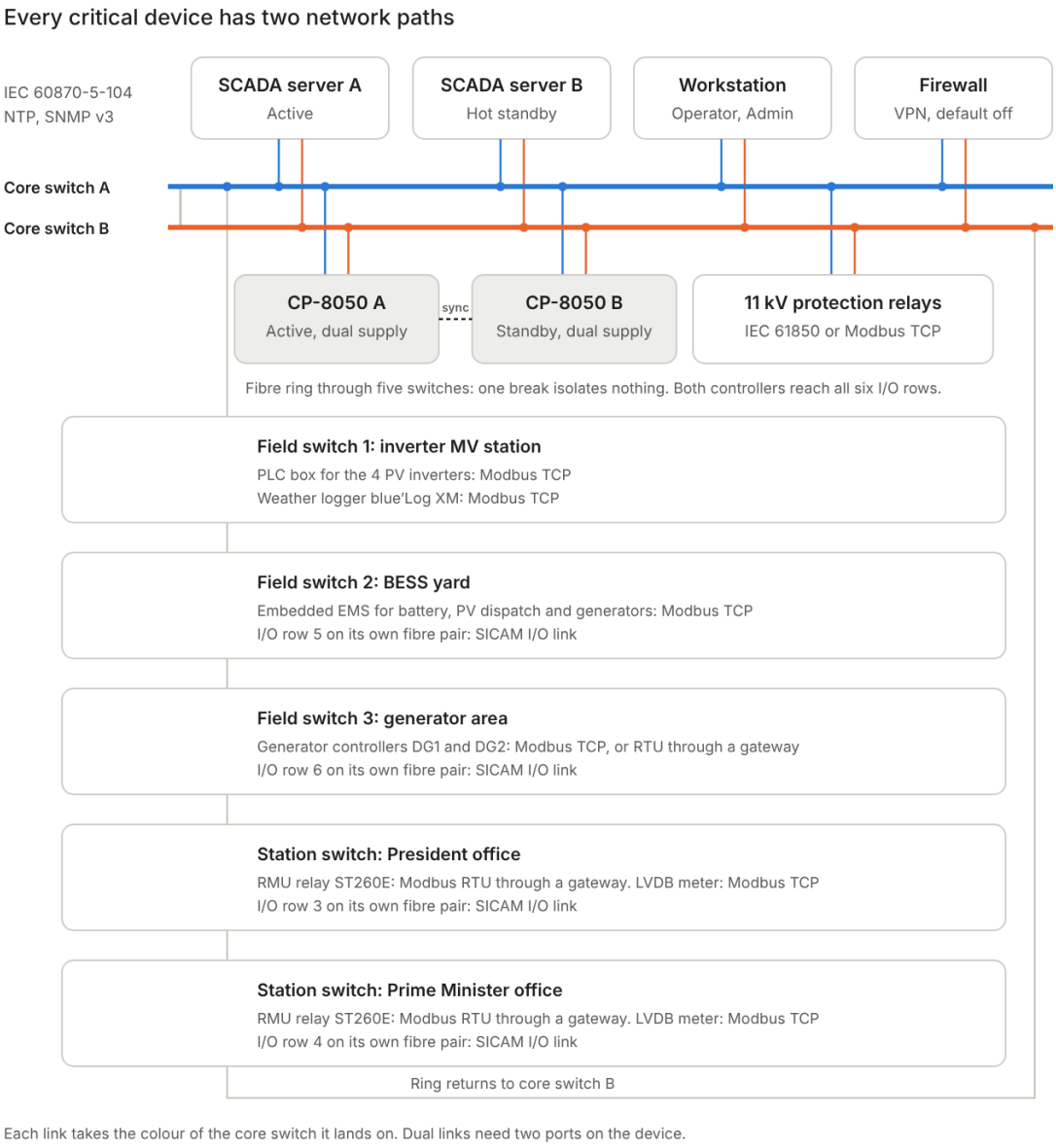
Source	Signals	Examples
Hardwired digital inputs	208	Breaker positions, trips, transformer alarms, fire panel states
Hardwired commands	32	Breaker open and close, alarm beacon
Hardwired analogue inputs	4	Generator fuel levels, DC bus

Source	Signals	Examples
		voltages
Communication points	463	Measurements, energy, engine values, battery data, settings

Breaker commands use the secured command module, which checks that only one output operates and that the coil circuit is intact before switching. Analogue inputs are used only for sensors without a communication port.

8. Communication architecture

All devices share one Ethernet network built on two core switches and a fibre ring. No single cable, switch, power supply or SCADA server failure removes the operators' view of the plant.



communication network · 2 core switches, 5-switch fibre ring, protocol per device

The SCADA servers, workstation, firewall, both controllers and the 11 kV relays each connect to both core switches. The field and station switches form one fibre ring that leaves core switch A and returns to core switch B.

Protocols by link

Link	Protocol	Medium	Redundant path
Controllers A and B to SCADA servers A and B	IEC 60870-5-104	Ethernet	Both core switches

Link	Protocol	Medium	Redundant path
Controller to 11 kV relays	IEC 61850, or Modbus TCP	Ethernet	Both core switches
PV inverters to PLC box	Power line communication	AC power cables	None needed; local
Controller to PLC box	Modbus TCP	Fibre ring	Ring
Controller to EMS (battery and generator data)	Modbus TCP	Fibre ring	Ring
Controller to generator controllers	Modbus TCP, or Modbus RTU through a gateway	Fibre ring	Ring
Controller to RMU relays (ST260E)	Modbus RTU through a station gateway	RS485, then fibre ring	Ring
Controller to power meters, weather logger, UPS	Modbus TCP or SNMP	Fibre ring or copper	Ring
Controllers to I/O rows 1 and 2	SICAM I/O link	Copper, in the cabinet	None; same cabinet
Controller to I/O rows 3 to 6	SICAM I/O link, optical	Dedicated fibre pair	None; see below
All devices to GPS time server	NTP	Ethernet	Both core switches
Network devices to SCADA servers	SNMP v3, syslog	Ethernet	Both core switches
Remote support	VPN through the firewall	Internet	None; off by default
Controller A to controller B	Redundancy synchronisation (Siemens)	Direct cable between the controllers, to confirm	Through the core switches, to confirm

Failure cases

Failure	Effect
One fibre cut in the ring	Traffic reroutes the other way; nothing is lost
One core switch fails	Dual-connected devices continue on the other

Failure	Effect
	switch
One field or station switch fails	Only devices on that switch are lost; the ring stays up
SCADA server A fails	Server B takes over with the same screens and history
One controller power supply fails	The second supply carries the controller
Mains supply to the SCADA cabinet fails	UPS and the station DC supply carry the SCADA and network
Active controller fails	The standby controller takes over; the SCADA view and remote switching continue; a level 2 alarm is raised
Fibre pair to a remote I/O row is cut	Hardwired signals from that location are lost until repaired; network data from that location still arrives through the ring
Both controllers fail	SCADA loses its view; the EMS, relays and hardwired trips continue; breakers are operated locally

Remaining single points

- **Remote I/O row links.** Each runs on one fibre pair. Siemens will confirm whether the optical row link can run as a ring; if not, a small CP-8031 controller at a site would put its signals on the ring.
- **The controller.** Controllers A and B run as a hot standby pair, so it is no longer a single point. Each I/O row interface remains single; its failure loses only that row.
- **Single-port devices.** The EMS, PLC box and generator controllers have one network port each, so they rely on the ring rather than on two connections.

9. Cybersecurity

The plant network is closed by default: no device can be reached from outside unless an Admin opens a time-limited VPN session. The provisions follow the structure of IEC 62443, and each layer assumes the one outside it might fail.

Network zones

Zone	Devices	Rule
SCADA	SCADA servers, workstation, engineering laptop	Reaches the controller only

Zone	Devices	Rule
Control	Controller, EMS	Only the controller writes settings to the EMS
Substation	Relays, power meters, gateways, I/O rows	Read and commanded by the controller only
Monitoring	PLC box, weather logger, UPS, time server	Read only
External	Firewall outside interface	Blocked unless an Admin opens a VPN session

Zones are separated by VLANs on the managed switches, by separate ports on the controller, and by firewall rules that deny all traffic not listed in section 8.

Provisions by layer

Layer	Provision
Perimeter	Firewall with deny-by-default rules; no port forwarding; no direct internet route to any control device
Remote access	VPN with multi-factor authentication; off by default; enabled per session by an Admin; closes on a timer; every session logged
Vendor access	Named, time-limited accounts through the same VPN; no permanent vendor access
Network	Managed switches with unused ports disabled, port security, encrypted management (HTTPS, SSH, SNMP v3)
Controller	Role-based access, encrypted engineering access, unused services disabled, event logs sent to the SCADA servers
SCADA	Individual logins, no shared accounts; minimum 12-character passwords; lockout after 5 failed attempts; logout after 15 idle minutes
Workstations	Application allow-listing, USB storage blocked, endpoint protection
Field devices	Default passwords changed on relays, EMS, PLC box, meters and generator controllers before the

Layer	Provision
	site test
Physical	Locked cabinets and control room; cabinet and panel door contacts raise alarms

Logging, time and recovery

- **Audit log.** Every login, command and setting change is recorded with user name and time, and kept for at least 12 months.
- **Time.** All devices synchronise to a GPS time server, so events from different devices can be put in order.
- **Backups.** Configurations of the controller, SCADA, switches, firewall and relays are backed up after every change, with one copy kept offline. Restore is tested at the site test.
- **Patching.** Firmware versions are frozen at the factory test. Updates are applied only after vendor release and Admin approval.
- **Recovery.** A written procedure covers local operation of breakers and generators if both controllers or both SCADA servers fail. If the EMS fails, the generators start on their own and supply the plant (section 11).

Decision for the client

No plant data leaves the site in the proposed design. If the client wants remote reporting or a vendor monitoring portal, it would need an outbound-only encrypted link, and the client should approve it explicitly given the nature of the site.

10. User roles and alarms

The SCADA has two user levels. An Operator runs the plant day to day; an Admin can also change how the plant behaves. Each person has an individual login, and every action is logged.

Permissions

Function	Operator	Admin
View all screens, trends, alarms and reports	Yes	Yes
Acknowledge alarms	Yes	Yes
Open and close breakers, with select and confirm	Yes	Yes
Start and stop a generator in manual mode	Yes	Yes
Clear level 3 (minor) faults and	Yes	Yes

Function	Operator	Admin
warnings		
Clear level 1 and level 2 faults	No	Yes
Change EMS operating mode and setpoints	No	Yes
Enable or disable the battery C-rate limit for calibration	No	Yes
Change generator mode between automatic and manual	No	Yes
Create and remove users, reset passwords	No	Yes
Enable a remote support session	No	Yes
Export the audit log	No	Yes

Settings that act on the battery depend on the EMS accepting them over Modbus. If the EMS does not expose the C-rate limit, that function stays on the EMS's own screen under its own Admin password.

Alarm levels

Level	Meaning	Examples	Cleared by
1, critical	Equipment tripped or safety at risk	Protection trip, fire, emergency stop, transformer trip, UPS battery low	Admin, after the cause is found; protection trips also need a reset at the relay
2, major	Function lost or degraded	Relay unhealthy, gas pressure low, generator alarm, communication lost	Admin
3, minor	Warning; the plant still runs normally	Temperature alarm, spring not charged, door open, surge arrester failed	Operator

The battery management system may number its own alarm levels differently. Its levels will be mapped to these three during configuration.

Breaker switching rules

1. Remote commands are accepted only when the breaker's local/remote switch is in remote.
2. The operator selects the breaker, then confirms in a second step.
3. The controller blocks a command that breaks an interlock, such as closing a breaker onto an earthed circuit, or energising the busbar through AH1 or AH3 while AH8 is open.
4. The secured command module checks that only one output operates and that the coil circuit is intact.
5. Before any work on equipment, the breaker position is checked physically, since screen status depends on communication.

11. Operational sequence

The battery forms the plant's voltage and frequency, PV follows it, and the generators start only when the battery runs low. The EMS carries out these sequences; the SCADA shows each step and lets authorised users intervene. Thresholds below are proposed starting values, to be set with the EMS supplier at commissioning.

Operating modes

Mode	Condition	Battery	PV	Generators
Day, PV surplus	PV exceeds load	Forms the grid, charging	Full output; curtailed by the EMS when the battery is full	Off
Day, PV shortfall	PV below load	Forms the grid, discharging	Full output	Off
Night	No PV	Forms the grid, discharging	Off	Off
Low battery	State of charge below 20%	Charging from the generator	Full output if available	One running; second starts if load requires
Generator only	Battery unavailable	Off	Limited to keep the generator above minimum load	Running, forming the grid
Manual	Admin selects manual	As commanded	As commanded	As commanded

Generator start and stop

1. State of charge falls below 20% and PV cannot cover the load.

2. The EMS requests a start from the lead generator.
3. The generator controller starts the engine, synchronises it to the busbar and closes its breaker.
4. The generator is loaded to at least 30% of its rating; spare output charges the battery.
5. If load exceeds 80% of the running generator, the second generator starts.
6. At 80% state of charge, or when PV covers the load again, the generator unloads and its breaker opens.
7. The engine cools down, then stops. The lead generator alternates to even out running hours.

The SCADA shows each step and raises an alarm if a generator fails to start, synchronise or stop.

Black start

1. After a total shutdown, all 11 kV breakers are open except AH8, which stays closed unless it has tripped.
2. AH8 is confirmed closed, or closed first, so the 11 kV system is earthed through the neutral earthing resistor before the busbar is energised. The controller blocks closing AH1 or AH3 while AH8 is open.
3. If the battery is healthy, it energises its transformer and the 11 kV busbar through AH3.
4. If the battery is not available, a generator energises the busbar through AH1.
5. The auxiliary transformer feeder AH5 closes, then the office feeders AH4 and AH6, with a delay between steps.
6. The PV incomer AH2 closes; the inverters reconnect and ramp up.

The EMS supplier provides this sequence in its functional description, states which steps it carries out automatically and which need an operator command from the SCADA, and demonstrates it at commissioning.

Requirements on the EMS and fallback

The design relies on the embedded EMS for dispatch, with no separate plant controller. The EMS supplier must therefore provide the functions below, and the plant must keep supplying the offices if the EMS fails. They are part of the EMS specification and are tested at commissioning.

- **Generator start and stop.** The EMS starts and stops each generator through a hardwired start contact to the generator controller, so the start does not depend on the CP-8050, the SCADA or the network. A Modbus command may duplicate it. The EMS applies the start below 20% and stop at 80% state of charge, the second-generator rule at 80% of the running generator, the 30% minimum load and lead and lag alternation.
- **PV curtailment.** The EMS limits the inverters through the PLC box when the battery is full, or when a generator would fall below its minimum load. Output is ramped down rather than tripped. The supplier states the response time from the EMS to the inverters.
- **Black start.** The EMS supplier provides the black-start sequence above in writing, including AH8 being closed first, and marks each step as automatic or by operator command.
- **EMS failure.** The CP-8050 monitors the EMS heartbeat and raises a level 1 alarm when it stops. Each generator controller is set to start on its own on a dead busbar, or when the EMS start

contact is lost, and to form the grid alone. The offices are then resupplied from the generators without the EMS, and the battery stays off until the EMS is restored.

- **Load shedding.** If a generator is overloaded and frequency falls, the 11 kV relays shed load by under-frequency protection (81): AH6, the Prime Minister feeder, first, and AH4, the President feeder, last. Proposed settings are 48.5 Hz for 1 s on AH6 and 47.5 Hz for 0.5 s on AH4, to be confirmed by the protection study. The auxiliary feeder AH5 is not shed. Feeders are reclosed by operator command once frequency is stable.

Fault responses

Event	Automatic response	Operator action
Protection trip on a feeder	Relay opens the breaker; level 1 alarm; remote reclose blocked	Find the cause; reset at the relay; Admin clears the alarm
Fire in the battery container	Hardwired trip of the battery; suppression releases; EMS starts a generator	Follow the site fire procedure; no remote reset
Emergency stop pressed	Hardwired stop of battery, PV and generators	Investigate; reset locally
Battery fault	EMS starts a generator and moves to generator-only mode	Call the Admin
Communication lost to a device	Level 2 alarm; values shown as stale; commands to that device blocked	Check the network; check breaker positions physically
Active controller failure	Standby controller takes over; level 2 alarm; SCADA view and remote switching continue	Repair or replace the failed controller; it rejoins as standby
SCADA server A failure	Server B takes over	Repair server A
EMS failure or EMS heartbeat lost	Level 1 alarm; generators start on dead busbar and supply the plant; battery off	Call the PV and BESS supplier; check generator loading
Generator overload, frequency falling	Relays shed AH6, then AH4, by under-frequency protection	Reduce load; reclose the feeders from the SCADA when frequency is stable
Failure of both controllers	Watchdog contact drops; alarm beacon sounds; EMS and relays continue	Operate locally under the recovery procedure

Battery C-rate calibration (Admin only)

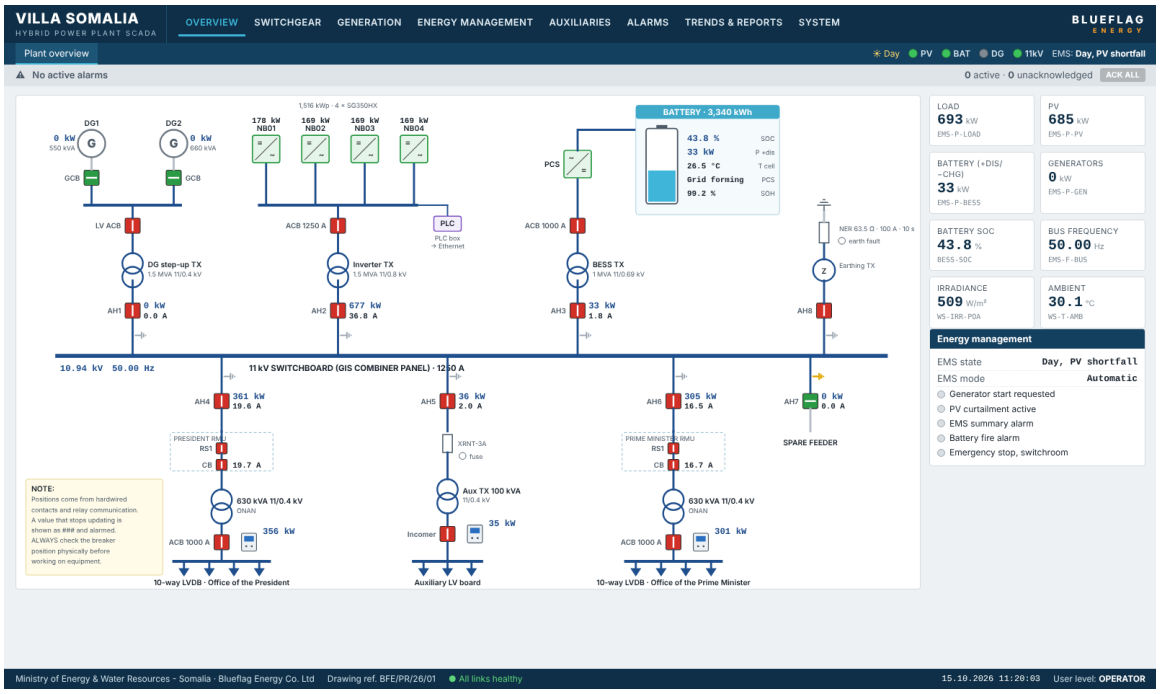
1. The Admin logs in and confirms a generator is available as backup.
2. The Admin disables the C-rate limit. The action is logged.
3. The calibration cycle runs under supervision.
4. The Admin re-enables the limit. An alarm reminds the Admin if it stays disabled beyond a set time.

12. HMI screens

The SCADA opens on a single-line overview of the whole plant; every other screen is one click away from it. Mock-ups of the main screens are shown below and the full set is in Attachment C. They are submitted for approval of layout and content; the final screens are built in SICAM SCC to the same layout before the factory test.

Screen	Content	Controls available
Plant overview	Live single-line diagram: generators, battery, PV, 11 kV busbar, office feeders; power at each point; breaker states; plant mode; summary alarms	Breaker open and close (Operator)
11 kV switchboard	Each panel AH1 to AH8: breaker and earth switch positions, local/remote, gas, relay status, currents, voltages, power, energy	Breaker open and close
Office stations	One screen per station: RMU positions, transformer alarms, LVDB incomer and 10 feeders, load and energy	RMU switches, transformer breaker, LV incomer
PV system	Four inverters: power, voltages, currents, temperature, insulation, state, fault codes; total PV output; weather data and performance ratio	None
Battery	State of charge and health, power, cell voltage and temperature limits, charge and discharge limits, alarms, fire panel state	EMS settings (Admin)
Generators	Each generator: electrical output, engine values, fuel	Start and stop in manual mode;

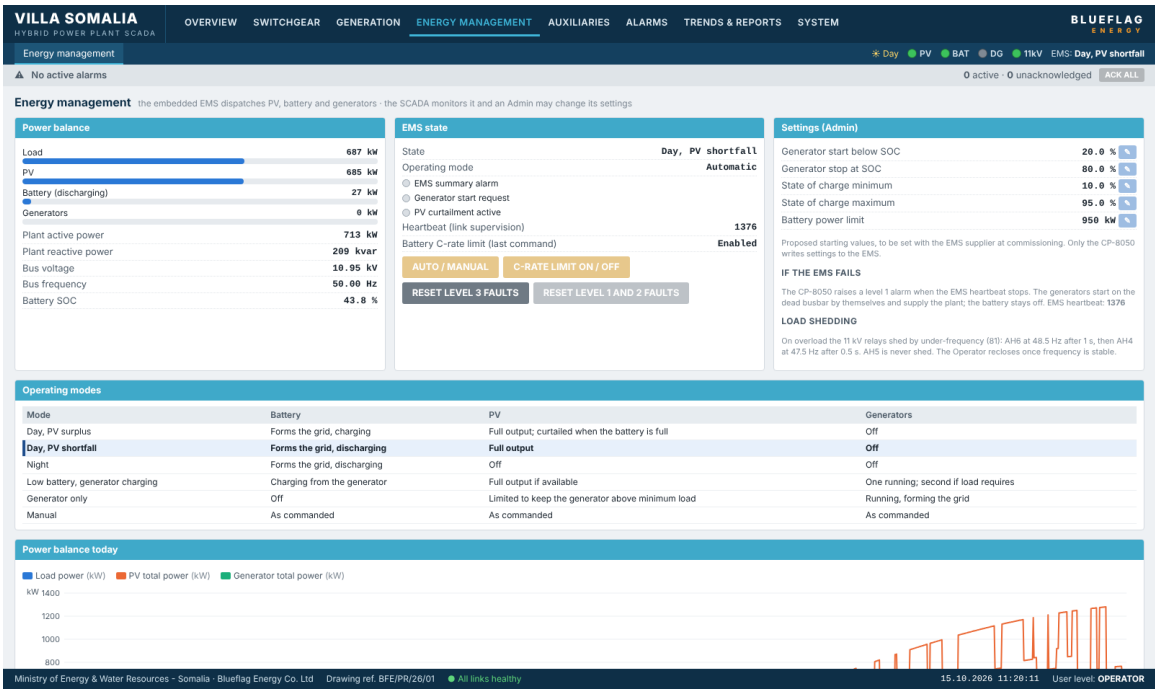
Screen	Content	Controls available
	level, running hours, starts, breaker, warnings and shutdowns	mode change (Admin)
Energy management	EMS mode, power balance, setpoints, generator start and stop thresholds	Mode and setpoints (Admin); C-rate limit (Admin)
Auxiliaries	Auxiliary transformer, UPS, DC supplies, switchroom temperature, doors	None
Alarms	Active alarms by level, with acknowledge; alarm history	Acknowledge; clear by level and role
Event log	Time-stamped record of every state change, command and login	Filter, export (Admin)
Trends	Any measured value over time	Select values and period
Reports	Daily, monthly and yearly energy by source, generator fuel and hours, availability	Export (Admin)
Communication status	Health of every device link, switch, I/O row and server	None
User management	Users and roles	Admin only
I/O list	Every signal with its tag, I/O row, module and channel or protocol source, alarm level and live value	Filter, export



Plant overview · mock-up of the opening screen, a live single-line diagram of the whole plant



11 kV switchboard · mock-up, panels AH1 to AH8 with positions, relay status and measurements



Energy management · mock-up, EMS state, power balance, operating modes and Admin settings

Screen conventions

- **Colours.** Breaker closed and open, alarm levels and stale values each use one consistent colour throughout.
- **Stale data.** A value from a device that has stopped communicating is greyed and marked, never shown as live.
- **Commands.** Every command opens a confirmation dialog naming the equipment and the action.
- **Role visibility.** Controls a user may not operate are hidden or disabled for that user's role.

13. Equipment and datasheets

The SCADA uses Siemens for the controller, I/O and SCADA software, and industrial-grade network and power equipment. The full bill of material, by panel and by order line, is Attachment D. Product numbers marked "to confirm" will be taken from the supplier quotations.

Controller and I/O (Siemens SICAM A8000)

Item	Model	Product number	Qty
Plant controller (processor module), hot standby pair	CP-8050	6MF2805-0AA00	2
Power supply, 100 to 240 V DC/AC, 45 W	PS-8642	6MF2864-2AA00	4

Item	Model	Product number	Qty
Ethernet interface, 5 ports	CI-8520	6MF2852-0AA00	2
Flash card, 2 GB	SICAM flash card	6MF1213-1GA05-0AA0	2
Digital input, 16 channels	DI-8112	6MF2811-2AA00	17
Secured command output	DO-8221	To confirm	4
Relay output, 8 channels	DO-8212	6MF2821-2AA00	1
Analogue input, 4 channels	AI-8320	6MF2832-0AA00	2
Remote I/O interface, electrical	CI-8532	6MF2853-2AA00	2
Remote I/O interface, optical	CI-8533	To confirm	4
Protocol licences: IEC 60870-5-104, Modbus TCP, IEC 61850, redundancy if licensed separately	For CP-8050	To confirm	2 sets

SCADA, network and power

Item	Manufacturer and model	Qty
SCADA servers, active and hot standby	Siemens SIMATIC IPC647E	2
SCADA software, redundant licence	Siemens SICAM SCC	1
Operator workstation with two monitors	Dell OptiPlex 7000 series	1
Engineering laptop with SICAM Device Manager	Dell Latitude 5430 Rugged	1
Core switches	Moxa EDS-G512E	2
Field and station switches	Moxa EDS-510E	5

Item	Manufacturer and model	Qty
Fibre transceivers	Moxa SFP-1GLXLC	14
Serial gateways for RMU relays	Moxa MGate MB3170	2
Firewall with VPN	Fortinet FortiGate Rugged 60F	1
GPS time server	Meinberg LANTIME M300	1
Power meters for LVDBs and auxiliary board	Schneider Electric PowerLogic PM5560	3
24 V DC power supplies	Phoenix Contact QUINT4-PS/1AC/24DC/20	6
UPS for the SCADA cabinet	APC Smart-UPS SRT 3000 VA	1
SCADA cabinet	Rittal VX25, 800 x 2000 x 800 mm	1
Outdoor I/O panels, IP55 minimum	Rittal AX	4
Alarm sounder and beacon	Pfannenberg PATROL	1
Armoured single-mode fibre, 12 core	To suit	3 routes
Network enclosure for SW-MV, IP55	Rittal AX	1
Server rack 24U with rack console	To suit	1

Spares

Item	Qty	Reason
Configured flash card for the CP-8050	1	Controller B is the running spare; the card restores a replaced controller quickly
DI-8112, DO-8221, CI-8533, PS-8642	1 each	One of each I/O module type and a controller power supply on site
Field switch and fibre transceivers	1 and 2	Network repair
QUINT4 24 V/20 A power supply	1 each	Field panel repair

Item	Qty	Reason
and MGate MB3170		

The controllers are supplied as a hot standby pair. I/O row 1 has its own CI-8532 interface rather than sitting on the controller backplane, so both controllers reach it. Siemens will confirm the redundancy function and licence, the switchover time, and how the I/O rows connect to both controllers.

Datasheets

- Siemens SICAM A8000: CP-8050, PS-8642, CI-8520, CI-8532, CI-8533, DI-8112, DO-8221, DO-8212, AI-8320
- Siemens SIMATIC IPC647E and SICAM SCC
- Moxa EDS-G512E, EDS-510E, SFP-1GLXLC, MGate MB3170
- Fortinet FortiGate Rugged 60F; Meinberg LANTIME M300
- Schneider Electric PM5560; Phoenix Contact QUINT4; APC Smart-UPS SRT
- Field devices from their suppliers: PLC box, EMS, generator controllers, 11 kV relays, ST260E, fire panel, UPS

14. Attachments

Attachment	Content	Status
A	SCADA architecture drawings BFE/PR/26/01-SCADA-001 to 005: architecture, communication network, PV interface, BESS interface and SCADA overview SLD (BFE-PR-26-01-SCADA-Drawing-Set_RevB.pdf)	Rev B, complete
B	Device and I/O list (Villa_Somalia_SCADA_Device_and_IO_List_Rev1.xlsx)	Rev 1, complete
C	HMI screen mock-ups: 12 screens from the HMI demonstration, including a battery fire (Villa_Somalia_SCADA_Attachment_C_HMI_Screens_Rev1.pdf)	Rev 1, complete; final screens built in SICAM SCC
D	Bill of material by panel and by order line, with I/O check and equipment by others (Villa_Somalia_SCADA_Bill_of_	Rev 1, for pricing

Attachment	Content	Status
	Material_Rev1.xlsx)	